

5.8.7

EE25BTECH11009 - Anshu kumar Ram

Question:

Yash scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each incorrect answer, then Yash would have scored 50 marks. How many questions were there in the test?

Solution:

Let the number of right and wrong answers be x and y respectively.

$$\begin{pmatrix} 3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 40 \quad (0.1)$$

$$\begin{pmatrix} 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 50 \quad (0.2)$$

These can be combined to give the matrix equation

$$\begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 40 \\ 50 \end{pmatrix} \quad (0.3)$$

This gives the augmented matrix

$$\left(\begin{array}{cc|c} 3 & -1 & 40 \\ 4 & -2 & 50 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - \frac{4}{3}R_1} \left(\begin{array}{cc|c} 3 & -1 & 40 \\ 0 & -\frac{2}{3} & -\frac{10}{3} \end{array} \right) \quad (0.4)$$

This gives the following values of x and y :

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 15 \\ 5 \end{pmatrix} \quad (0.5)$$

Thus, the number of questions in the test is

$$x + y = 20 \quad (0.6)$$

